# Adding Integers

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# Integers

**Integers** are whole numbers, their opposites, and zero.

They include positive integers, negative integers, and zero.

$$\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$$

# Positive Integers

**Positive integers** are integers greater than zero.

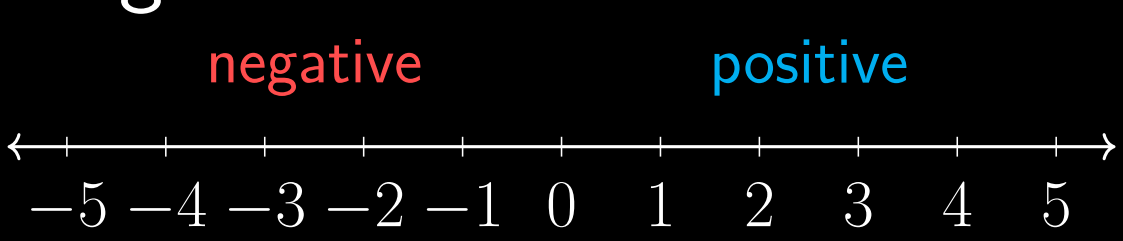
$$1,\ 2,\ 3,\ \ldots$$

# Negative Integers

**Negative integers** are integers less than zero.

$$-1,\ -2,\ -3,\ \ldots$$

# Integers on a Number Line

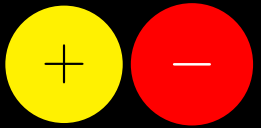



Right of zero: positive.

Left of zero: negative.

# Zero Pairs

A **zero pair** is one positive counter and one negative counter.




$$= 0$$

$$1 + (-1) = 0$$

# Adding with a Number Line

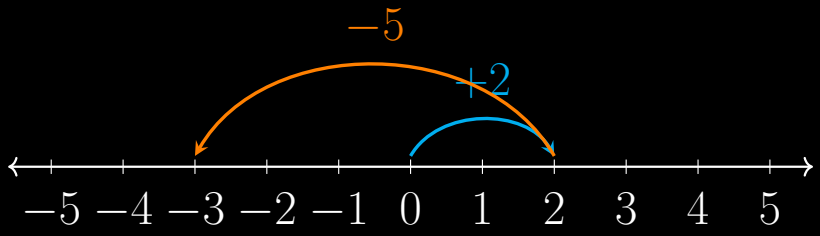
1. Start at $0$.
2. Move for the first integer.
3. Move for the second integer.
4. The final position is the sum.

# Number Line Directions

| **Integer is...** | **Move...** |
| --- | --- |
| Positive | Right $\rightarrow$ |
| Negative | Left $\leftarrow$ |

# Number Line Example

$$2 + (-5) = -3$$




Start at $0$, move $2$ right, then $5$ left.

# Same Signs

**Adding integers with the same signs:**

Add their absolute values and keep their common sign.

$$4 + 3 = 7$$

$$(-4) + (-3) = -7$$

# Different Signs

**Adding integers with different signs:**

Subtract the smaller absolute value from the larger absolute value.

Use the sign of the integer with the greater absolute value.

# Different Signs: Examples

$$7 + (-3) = 4$$

$7$ has the greater absolute value, so the answer is positive.

$$(-8) + 5 = -3$$

$-8$ has the greater absolute value, so the answer is negative.

# Adding Zero

**Adding zero** to any integer does not change its value.

$$(-6) + 0 = -6$$

# Quick Guide

- **Signs: Same** $\rightarrow$ Add, keep the sign
- **Signs: Different** $\rightarrow$ Subtract, use the sign of the bigger one

# Remember

**Same signs: add and keep.**
**Different signs: subtract, and the bigger one wins the sign.**

# Examples

# Example 1

Find $4 + 3$.

Same signs: add and keep the sign.

$$4 + 3 = 7$$

# Example 2

Find $(-4) + (-3)$.

Same signs: add $4 + 3 = 7$ and keep the negative sign.

$$(-4) + (-3) = -7$$

# Example 3

Find $7 + (-3)$.

Different signs: $7-3 = 4$. The larger absolute value is $7$, which is positive.

$$7 + (-3) = 4$$

# Example 4

Find $(-8) + 5$.

Different signs: $8 - 5 = 3$. The larger absolute value is $8$, which is negative.

$$(-8) + 5 = -3$$

## Example 5

Find $(-6) + 0$.

Adding zero does not change the value.

$$(-6) + 0 = -6$$

# Practice Exercises

Practice 1

Find the sum.

$$4 + (-2)$$

Practice 2

Find the sum.

$$(-5) + 3$$

Practice 3

Find the sum.

$$(-3) + (-4)$$

Practice 4

Find the sum.

$$6 + (-6)$$

Practice 5

Find the sum.

$$2 + 4$$

Practice 6

Find the sum.

$$(-3)+5$$

Practice 7

Find the sum.

$$4 + (-7)$$

Practice 8

Find the sum.

$$8 + 6$$

Practice 9

Find the sum.

$$(-7) + (-5)$$

Practice 10

Find the sum.

$$(-9) + 4$$

# Activities

Activity 1

Find the sum.

$$5 + (-3)$$

Activity 2

# Find the sum.

$$(-6) + 4$$

Activity 3

Find the sum.

$$(-2) + (-5)$$

Activity 4

Find the sum.

$$7 + (-7)$$

Activity 5

# Find the sum.

$$3 + 5$$

Activity 6

# Find the sum.

$$(-4) + 6$$

Activity 7

# Find the sum.

$$5 + (-8)$$

Activity 8

# Find the sum.

$$9 + 7$$

Activity 9

Find the sum.

$$(-8) + (-6)$$

Activity 10

Find the sum.

$$(-10) + 3$$

# Thank you!